

PREDICTING A POPULATION'S SIZE AND ITS CHANCES OF EXTINCTION

POPULATION VIABILITY ANALYSIS

IN CONTEXT

KEY FIGURES

Mark L. Shaffer (1949–) **Georgina Mace** (1953–2020)

BEFORE

1964 The IUCN publishes its first Red List of threatened mammal and bird species.

1965 In *The Destruction of California*, ecologist Raymond Dasmann charts the rapid loss of flora and fauna in the state.

1967 *The Theory of Island Biogeography* by Robert MacArthur and Edward O. Wilson explores island patterns of immigration and extinction.

AFTER

2003 Population viability analysis (PVA) of the Fender's blue butterfly is used to guide conservation in the US.

2014 PVA studies in the Sonoran Desert assess bird and reptile response to climate change.

Population viability analysis (PVA), or extinction risk assessment, is a process used to estimate the probability that a population of a target species has the ability to sustain itself for a specific time, be it 10, 30, or 100 years.